1.1 Basic Constructions

This first section begins with the basic definitions and constructions, and then proceeds quickly to an important calculation, the fundamental group of the circle, using notions developed more fully in §1.3. More systematic methods of calculation are given in §1.2. These are sufficient to show for example that every group is realized as the fundamental group of some space. This idea is exploited in the Additional Topics at the end of the chapter, which give some illustrations of how algebraic facts about groups can be derived topologically, such as the fact that every subgroup of a free group is free.

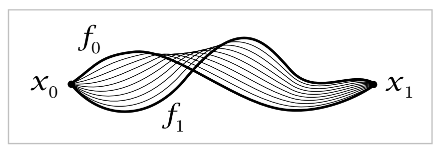
Paths and Homotopy

The fundamental group will be defined in terms of loops and deformations of loops. Sometimes it will be useful to consider more generally paths and their deformations, so we begin with this slight extra generality.

By a **path** in a space X we mean a continuous map $f: I \to X$ where I is the unit interval $[0,1]$. The idea of continuously deforming a path, keeping its endpoints fixed, is made precise by the following definition. A **homotopy** of paths in X is a family $f_t: I \to X$, $0 \le t \le 1$, such that

(1) The endpoints $f_t(0) = x_0$ and $f_t(1) = x_1$ are independent of t.

(2) The associated map $F: I \times I \to X$ defined by $F(s,t) = f_t(s)$ is continuous.



When two paths f_0 and f_1 are connected in this way by a homotopy f_t, they are said to be **homotopic**. The notation for this is $f_0 \simeq f_1$.

Example 1.1: Linear Homotopies. Any two paths f_0 and f_1 in $\mathbb{R}^n$ having the same endpoints x_0 and x_1 are homotopic via the homotopy $f_t(s) = (1-t)f_0(s) + tf_1(s)$. During this homotopy each point $f_0(s)$ travels along the line segment to $f_1(s)$ at constant speed. This is because the line through $f_0(s)$ and $f_1(s)$ is linearly parametrized as $f_0(s) + t[f_1(s) - f_0(s)] = (1-t)f_0(s) + tf_1(s)$, with the segment from $f_0(s)$ to $f_1(s)$ covered by t values in the interval from 0 to 1. If $f_1(s)$ happens to equal $f_0(s)$ then this segment degenerates to a point and $f_t(s) = f_0(s)$ for all t. This occurs in particular for $s = 0$ and $s = 1$, so each f_t is a path from x_0 to x_1. Continuity of the homotopy f_t as a map $I \times I \to \mathbb{R}^n$ follows from continuity of f_0 and f_1 since the algebraic operations of vector addition and scalar multiplication in the formula for f_t are continuous.

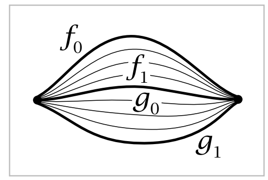
This construction shows more generally that for a convex subspace $X \subset \mathbb{R}^n$, all paths in X with given endpoints x_0 and x_1 are homotopic, since if f_0 and f_1 lie in X then so does the homotopy f_t.

Before proceeding further we need to verify a technical property:

Proposition 1.2. *The relation of homotopy on paths with fixed endpoints in any space is an equivalence relation.*

The equivalence class of a path f under the equivalence relation of homotopy will be denoted $[f]$ and called the **homotopy class** of f.

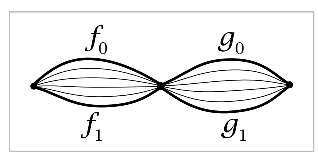
Proof: Reflexivity is evident since $f \simeq f$ by the constant homotopy $f_t = f$. Symmetry is also easy since if $f_0 \simeq f_1$ via f_t, then $f_1 \simeq f_0$ via the inverse homotopy f_{1-t}. For transitivity, if $f_0 \simeq f_1$ via f_t and if $f_1 = g_0$ with $g_0 \simeq g_1$ via g_t, then $f_0 \simeq g_1$ via the homotopy h_t that equals f_{2t} for $0 \le t \le {}^1\!/_2$ and g_{2t-1} for ${}^1\!/_2 \le t \le 1$. These two definitions agree for $t = {}^1\!/_2$ since we assume $f_1 = g_0$. Continuity of the associated map $H(s,t) = h_t(s)$ comes from the elementary fact, which will be used frequently without explicit mention, that a function defined on the union of two closed sets is continuous if it is continuous when restricted to each of the closed sets separately. In the case at hand we have $H(s,t) = F(s,2t)$ for $0 \le t \le {}^1\!/_2$ and $H(s,t) = G(s,2t-1)$ for ${}^1\!/_2 \le t \le 1$ where F and G are the maps $I \times I \to X$ associated to the homotopies f_t and g_t. Since H is continuous on $I \times [0, {}^1\!/_2]$ and on $I \times [{}^1\!/_2, 1]$, it is continuous on $I \times I$. $\square$



Given two paths $f, g : I \to X$ such that $f(1) = g(0)$, there is a **composition** or **product path** $f \cdot g$ that traverses first f and then g, defined by the formula

$$f \cdot g(s) = \begin{cases} f(2s), & 0 \le s \le {}^1\!/_2 \\ g(2s-1), & {}^1\!/_2 \le s \le 1 \end{cases}$$

Thus f and g are traversed twice as fast in order for $f \cdot g$ to be traversed in unit time. This product operation respects homotopy classes since if $f_0 \simeq f_1$ and $g_0 \simeq g_1$ via homotopies f_t and g_t, and if $f_0(1) = g_0(0)$ so that $f_0 \cdot g_0$ is defined, then $f_t \cdot g_t$ is defined and provides a homotopy $f_0 \cdot g_0 \simeq f_1 \cdot g_1$.



In particular, suppose we restrict attention to paths $f : I \to X$ with the same starting and ending point $f(0) = f(1) = x_0 \in X$. Such paths are called **loops**, and the common starting and ending point x_0 is referred to as the **basepoint**. The set of all homotopy classes $[f]$ of loops $f : I \to X$ at the basepoint x_0 is denoted $\pi_1(X, x_0)$.

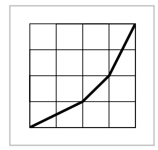
Proposition 1.3. $\pi_1(X, x_0)$ *is a group with respect to the product* $[f][g] = [f \cdot g]$.

This group is called the **fundamental group** of X at the basepoint x_0. We will see in Chapter 4 that $\pi_1(X, x_0)$ is the first in a sequence of groups $\pi_n(X, x_0)$, called homotopy groups, which are defined in an entirely analogous fashion using the n-dimensional cube I^n in place of I.

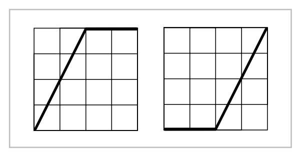
Proof: By restricting attention to loops with a fixed basepoint $x_0 \in X$ we guarantee that the product $f \cdot g$ of any two such loops is defined. We have already observed that the homotopy class of $f \cdot g$ depends only on the homotopy classes of f and g, so the product $[f][g] = [f \cdot g]$ is well-defined. It remains to verify the three axioms for a group.

As a preliminary step, define a **reparametrization** of a path f to be a composition $f\varphi$ where $\varphi : I \to I$ is any continuous map such that $\varphi(0) = 0$ and $\varphi(1) = 1$. Reparametrizing a path preserves its homotopy class since $f\varphi \simeq f$ via the homotopy $f\varphi_t$ where $\varphi_t(s) = (1-t)\varphi(s) + ts$ so that $\varphi_0 = \varphi$ and $\varphi_1(s) = s$. Note that $(1-t)\varphi(s) + ts$ lies between $\varphi(s)$ and s, hence is in I, so the composition $f\varphi_t$ is defined.

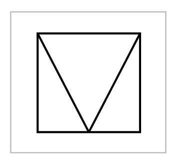
If we are given paths f, g, h with $f(1) = g(0)$ and $g(1) = h(0)$, then both products $(f \cdot g) \cdot h$ and $f \cdot (g \cdot h)$ are defined, and $f \cdot (g \cdot h)$ is a reparametrization of $(f \cdot g) \cdot h$ by the piecewise linear function φ whose graph is shown in the figure at the right. Hence $(f \cdot g) \cdot h \simeq f \cdot (g \cdot h)$. Restricting attention to loops at the basepoint x_0, this says the product in $\pi_1(X, x_0)$ is associative.



Given a path $f : I \to X$, let c be the constant path at $f(1)$, defined by $c(s) = f(1)$ for all $s \in I$. Then $f \cdot c$ is a reparametrization of f via the function φ whose graph is shown in the first figure at the right, so $f \cdot c \simeq f$. Similarly, $c \cdot f \simeq f$ where c is now the constant path at $f(0)$, using the reparametrization function in the second figure. Taking f to be a loop, we deduce that the homotopy class of the constant path at x_0 is a two-sided identity in $\pi_1(X, x_0)$.



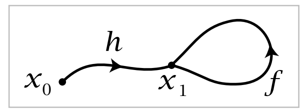
For a path f from x_0 to x_1, the **inverse path** $\overline{f}$ from x_1 back to x_0 is defined by $\overline{f}(s) = f(1-s)$. To see that $f \cdot \overline{f}$ is homotopic to a constant path we use the homotopy $h_t = f_t \cdot g_t$ where f_t is the path that equals f on the interval $[0, 1-t]$ and that is stationary at $f(1-t)$ on the interval $[1-t, 1]$, and g_t is the inverse path of f_t. We could also describe h_t in terms of the associated function $H : I \times I \to X$ using the decomposition of $I \times I$ shown in the figure. On the bottom edge of the square H is given by $f \cdot \overline{f}$ and below the 'V' we let $H(s,t)$ be independent of t, while above the 'V' we let $H(s,t)$ be independent of s. Going back to the first description of h_t, we see that since $f_0 = f$ and f_1 is the constant path c at x_0, h_t is a homotopy from $f \cdot \overline{f}$ to $c \cdot \overline{c} = c$. Replacing f by $\overline{f}$ gives $\overline{f} \cdot f \simeq c$ for c the constant path at x_1. Taking f to be a loop at the basepoint x_0, we deduce that $[\,\overline{f}\,]$ is a two-sided inverse for $[f]$ in $\pi_1(X, x_0)$. $\square$



Example 1.4. For a convex set X in $\mathbb{R}^n$ with basepoint $x_0 \in X$ we have $\pi_1(X, x_0) = 0$, the trivial group, since any two loops f_0 and f_1 based at x_0 are homotopic via the linear homotopy $f_t(s) = (1-t)f_0(s) + tf_1(s)$, as described in Example 1.1.

It is not so easy to show that a space has a nontrivial fundamental group since one must somehow demonstrate the nonexistence of homotopies between certain loops. We will tackle the simplest example shortly, computing the fundamental group of the circle.

It is natural to ask about the dependence of $\pi_1(X,x_0)$ on the choice of the basepoint x_0. Since $\pi_1(X,x_0)$ involves only the path-component of X containing x_0, it is clear that we can hope to find a relation between $\pi_1(X,x_0)$ and $\pi_1(X,x_1)$ for two basepoints x_0 and x_1 only if x_0 and x_1 lie in the same path-component of X. So let $h:I\to X$ be a path from x_0 to x_1, with the inverse path $\overline{h}(s)=h(1-s)$ from x_1 back to x_0. We can then associate to each loop f based at x_1 the loop $h\cdot f\cdot\overline{h}$ based at x_0.



Strictly speaking, we should choose an order of forming the product $h\cdot f\cdot\overline{h}$, either $(h\cdot f)\cdot\overline{h}$ or $h\cdot(f\cdot\overline{h})$, but the two choices are homotopic and we are only interested in homotopy classes here. Alternatively, to avoid any ambiguity we could define a general n-fold product $f_1\cdot\,\cdots\,\cdot f_n$ in which the path f_i is traversed in the time interval $[\frac{i-1}{n},\frac{i}{n}]$.

Proposition 1.5. *The map $\beta_h:\pi_1(X,x_1)\to\pi_1(X,x_0)$ defined by $\beta_h[f]=[h\cdot f\cdot\overline{h}]$ is an isomorphism.*

Proof: If f_t is a homotopy of loops based at x_1 then $h\cdot f_t\cdot\overline{h}$ is a homotopy of loops based at x_0, so β_h is well-defined. Further, β_h is a homomorphism since $\beta_h[f\cdot g]=[h\cdot f\cdot g\cdot\overline{h}]=[h\cdot f\cdot\overline{h}\cdot h\cdot g\cdot\overline{h}]=\beta_h[f]\beta_h[g]$. Finally, β_h is an isomorphism with inverse $\beta_{\overline{h}}$ since $\beta_h\beta_{\overline{h}}[f]=\beta_h[\overline{h}\cdot f\cdot h]=[h\cdot\overline{h}\cdot f\cdot h\cdot\overline{h}]=[f]$, and similarly $\beta_{\overline{h}}\beta_h[f]=[f]$. $\square$

Thus if X is path-connected, the group $\pi_1(X,x_0)$ is, up to isomorphism, independent of the choice of basepoint x_0. In this case the notation $\pi_1(X,x_0)$ is often abbreviated to $\pi_1(X)$, or one could go further and write just π_1X.

In general, a space is called **simply-connected** if it is path-connected and has trivial fundamental group. The following result explains the name.

Proposition 1.6. *A space X is simply-connected iff there is a unique homotopy class of paths connecting any two points in X.*

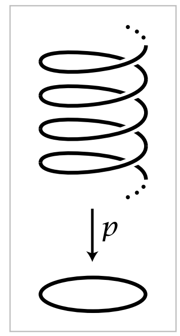
Proof: Path-connectedness is the existence of paths connecting every pair of points, so we need be concerned only with the uniqueness of connecting paths. Suppose $\pi_1(X)=0$. If f and g are two paths from x_0 to x_1, then $f\simeq f\cdot\overline{g}\cdot g\simeq g$ since the loops $\overline{g}\cdot g$ and $f\cdot\overline{g}$ are each homotopic to constant loops, using the assumption $\pi_1(X,x_0)=0$ in the latter case. Conversely, if there is only one homotopy class of paths connecting a basepoint x_0 to itself, then all loops at x_0 are homotopic to the constant loop and $\pi_1(X,x_0)=0$. $\square$

The Fundamental Group of the Circle

Our first real theorem will be the calculation $\pi_1(S^1) \approx \mathbb{Z}$. Besides its intrinsic interest, this basic result will have several immediate applications of some substance, and it will be the starting point for many more calculations in the next section. It should be no surprise then that the proof will involve some genuine work. To maximize the payoff for this work, the proof is written so that its main technical steps apply in the more general setting of covering spaces, the main topic of §1.3.

Theorem 1.7. *The map $\Phi : \mathbb{Z} \to \pi_1(S^1)$ sending an integer n to the homotopy class of the loop $\omega_n(s) = (\cos 2\pi ns, \sin 2\pi ns)$ based at $(1,0)$ is an isomorphism.*

Proof: The idea is to compare paths in S^1 with paths in $\mathbb{R}$ via the map $p : \mathbb{R} \to S^1$ given by $p(s) = (\cos 2\pi s, \sin 2\pi s)$. This map can be visualized geometrically by embedding $\mathbb{R}$ in $\mathbb{R}^3$ as the helix parametrized by $s \mapsto (\cos 2\pi s, \sin 2\pi s, s)$, and then p is the restriction to the helix of the projection of $\mathbb{R}^3$ onto $\mathbb{R}^2$, $(x,y,z) \mapsto (x,y)$, as in the figure. Observe that the loop ω_n is the composition $p\widetilde{\omega}_n$ where $\widetilde{\omega}_n : I \to \mathbb{R}$ is the path $\widetilde{\omega}_n(s) = ns$, starting at 0 and ending at n, winding around the helix $|n|$ times, upward if $n > 0$ and downward if $n < 0$. The relation $\omega_n = p\widetilde{\omega}_n$ is expressed by saying that $\widetilde{\omega}_n$ is a **lift** of ω_n.



The definition of Φ can be reformulated by setting $\Phi(n)$ equal to the homotopy class of the loop $p\widetilde{f}$ for $\widetilde{f}$ any path in $\mathbb{R}$ from 0 to n. Such an $\widetilde{f}$ is homotopic to $\widetilde{\omega}_n$ via the linear homotopy $(1-t)\widetilde{f} + t\widetilde{\omega}_n$, hence $p\widetilde{f}$ is homotopic to $p\widetilde{\omega}_n = \omega_n$ and the new definition of $\Phi(n)$ agrees with the old one.

To verify that Φ is a homomorphism, let $\tau_m : \mathbb{R} \to \mathbb{R}$ be the translation $\tau_m(x) = x + m$. Then $\widetilde{\omega}_m \cdot (\tau_m \widetilde{\omega}_n)$ is a path in $\mathbb{R}$ from 0 to $m+n$, so $\Phi(m+n)$ is the homotopy class of the loop in S^1 that is the image of this path under p. This image is just $\omega_m \cdot \omega_n$, so $\Phi(m+n) = \Phi(m) \cdot \Phi(n)$.

To show that Φ is an isomorphism we shall use two facts:

(a) For each path $f : I \to S^1$ starting at a point $x_0 \in S^1$ and each $\widetilde{x}_0 \in p^{-1}(x_0)$ there is a unique lift $\widetilde{f} : I \to \mathbb{R}$ starting at $\widetilde{x}_0$.

(b) For each homotopy $f_t : I \to S^1$ of paths starting at x_0 and each $\widetilde{x}_0 \in p^{-1}(x_0)$ there is a unique lifted homotopy $\widetilde{f}_t : I \to \mathbb{R}$ of paths starting at $\widetilde{x}_0$.

Before proving these facts, let us see how they imply the theorem. To show that Φ is surjective, let $f : I \to S^1$ be a loop at the basepoint $(1,0)$, representing a given element of $\pi_1(S^1)$. By (a) there is a lift $\widetilde{f}$ starting at 0. This path $\widetilde{f}$ ends at some integer n since $p\widetilde{f}(1) = f(1) = (1,0)$ and $p^{-1}(1,0) = \mathbb{Z} \subset \mathbb{R}$. By the extended definition of Φ we then have $\Phi(n) = [p\widetilde{f}] = [f]$. Hence Φ is surjective.